

(a) Text-Guided Semantic Filtering

Description Generation



What does a [CLASS] should look like?



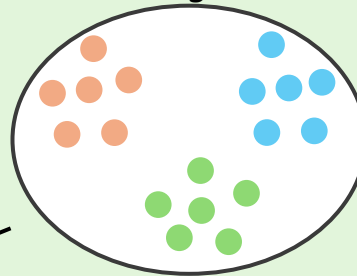
It has a soft coat, floppy ears, and a wagging tail.

Class Descriptions

- An aircraft has wide fixed wings, engines, and a streamlined fuselage.
- ...
- Dog has a soft coat, floppy ears, and a wagging tail.

text encoder

Knowledge Base

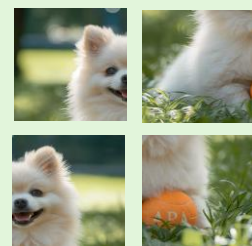
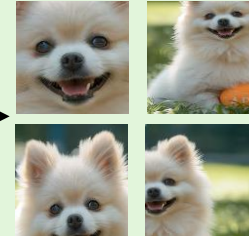


Text Anchor guided selection

Top $q\%$

- | | | | | |
|----|---|---|------------------------|---|
| ★1 | ↔ | { | Align.: 0.9 Conf.: 0.8 | ✓ |
| ★2 | ↔ | { | Align.: 0.6 Conf.: 0.8 | ✓ |
| ★3 | ↔ | { | Align.: 0.7 Conf.: 0.5 | ✓ |
| ⋮ | | | | |
| ★N | ↔ | { | Align.: 0.3 Conf.: 0.2 | ✗ |

Text-guided Selected Views



N Augmented Views

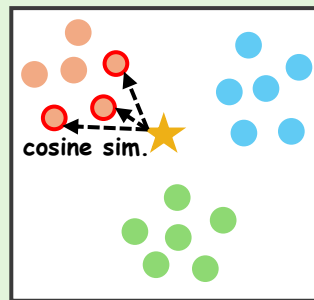
image encoder

★1

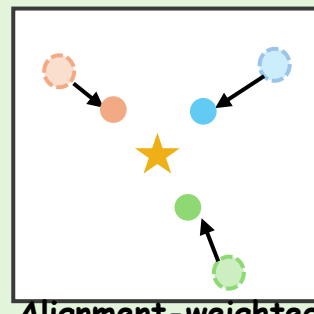
⋮

★N

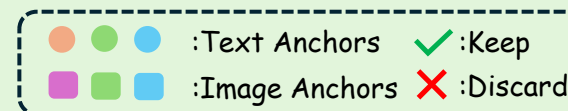
Image embeds



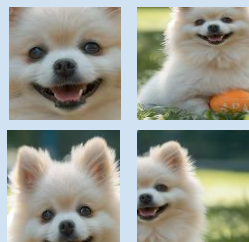
Compute alignment score



Alignment-weighted aggregation



(b) Image-Guided Semantic Filtering



Text-guided Selected Views

image encoder

★

⋮

★

Image embeds

update

Prototype Bank

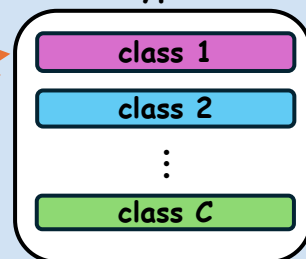
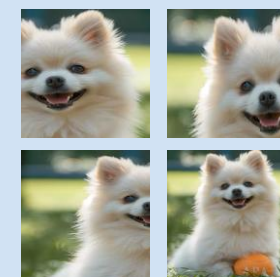


Image Anchor guided selection

Top $p\%$

- | | | | | |
|----|---|---|------------------------|---|
| ★1 | ↔ | { | Align.: 0.9 Conf.: 0.7 | ✓ |
| ★2 | ↔ | { | Align.: 0.8 Conf.: 0.6 | ✓ |
| ★3 | ↔ | { | Align.: 0.7 Conf.: 0.6 | ✓ |
| ⋮ | | | | |
| ★N | ↔ | { | Align.: 0.2 Conf.: 0.2 | ✗ |

Image-guided Selected Views



Union

Final Set